UNIT 2: [Name of the unit]

Chapter 01: [Robotics domain]
Lesson 01: [Detect the fire] Time Frame: [90 min]

Objectives:

1)understand of relay and how to use it with water pump.

2)understand how the flame sensor detect the fire.

Challenge questions of the lesson:

- How can we design a system to control a fire truck and effectively put out fires?
- What is the best way to detect a fire quickly and alert people?
- How can we combine control systems and detection systems to respond to emergencies?

SEL Relation

- **Know Yourself:**
  - Know what you are good at and what is hard.
  - Know how you feel (happy, mad, etc.).
- **Manage Yourself:**
  - Handle feelings (like not getting too mad).
  - Finish your work.
  - Keep trying when it's hard.
- **Know About Others:**
  - Listen to friends' ideas.
  - Know how to work in a group.
- **Work with Others:**
  - Talk nicely to your group.
  - Help your friends build and fix things.
  - Be a good helper in the group.
- **Make Good Choices:**
  - Find problems and fix them.
  - Think about the best way to build or fix something.
  - Be safe with tools and water.

Lesson Plan

1. Engage

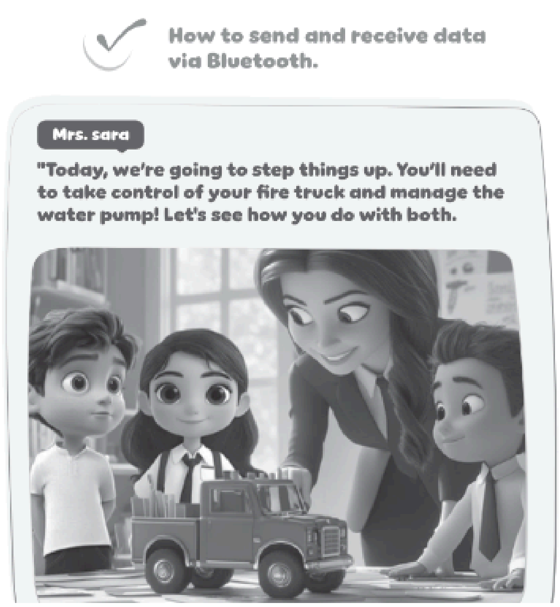
Teacher Role:

🎬 **Facilitator:** Guide the discussion and activities without giving away the solutions.
🎬 **Questioner:** Ask open-ended questions to probe student thinking and encourage participation.
🎬 **Motivator:** Create excitement and relevance around the topic of fire safety and emergency response systems.
🎬 **Listener:** Pay close attention to student responses to gauge their understanding and interests.

Expected From Students:

🎬 **Active Participation:** Share ideas, ask questions, and respond to prompts.
🎬 **Brainstorming:** Think about fire trucks, water pumps, fire detection, and alarms.
🎬 **Making Connections:** Relate the topic to their own experiences or prior knowledge.
🎬 **Expressing Curiosity:** Show interest in the challenge presented by Mrs. Sara and Laila's suggestion.
🎬 **Predicting/Hypothesizing (Initial):** Start thinking about how controlling a fire truck or detecting fire might work.

Parts of the books included.



- **Strategies implemented:**

Visual or Multimedia Engagement:

- Show a short, engaging video clip of a fire truck in action (responding to a call, using its water cannon).
- Show images of different types of fire detection systems (smoke detectors, heat sensors).
- Use visuals to help students visualize the concepts discussed in the story snippet

2. Think

- **Questions**

  Why it is necessary to control the water pump?

  List ways of how can we control the water pump?

- **Possible Answers**
  Possible Student Answers: To put out the fire effectively; to save water; to adjust the water pressure; to aim the water correctly; to avoid damaging things with too much water pressure; to make sure the water lasts longer.
  Possible Student Answers: With buttons; with levers; with a computer or screen; with a remote control; automatically (like a sensor); with a steering wheel or joystick; by turning a handle.

- **Strategies to implement:**
  1.(Think-Pair-Share): have students think individually first, then discuss in small groups (Think-Pair-Share). Record student ideas on a whiteboard or chart paper. Encourage diverse answers and probe their reasoning.
  2.brainstorming: Facilitate a brainstorming session. Ask students to think about different devices they use that have controls (video games, cars, appliances). Encourage creative ideas. Group similar ideas together.

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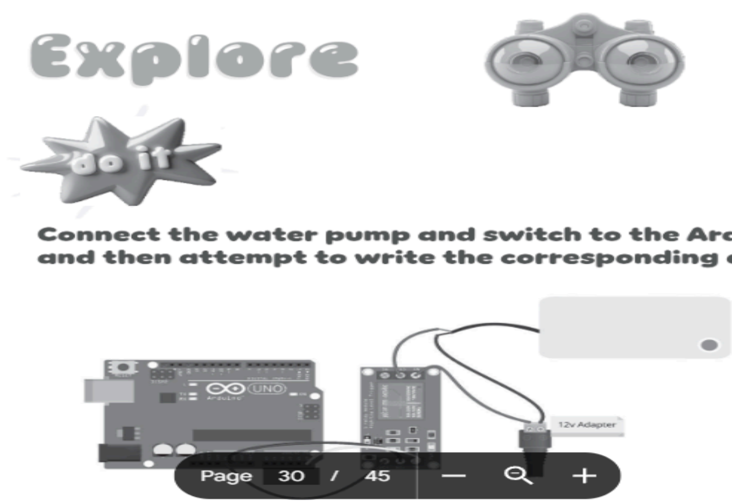
3. Explore

- **Teacher Role:**
  Facilitator: Guide students through the activity, providing support and hints without giving direct answers.
  Observer: Circulate among groups, observe student interactions, troubleshooting processes, and understanding.
  Questioner: Ask guiding questions to help students think through challenges and understand the concepts (e.g., "What happens when you connect this wire here?", "What does this line of code do?").
  Safety Monitor: Ensure students handle components and water safely.
  Resource Provider: Make sure all groups have the necessary materials and access to computers with the Arduino IDE.

- **Expected From Students:**
  Collaborate: Work effectively in pairs or small groups.
  Build Circuits: Follow diagrams or instructions to connect the switch, water pump (via a relay), and Arduino.
  Write/Modify Code: Attempt to write or modify basic Arduino code to read the switch input and control the relay/pump output.
  Test and Troubleshoot: Test their circuit and code, identify errors, and attempt to fix them.
  Record Observations: Note what happens when they interact with the circuit and code.
  Experiment: Try different connections or code modifications (if time allows and is appropriate).

- **Parts of the books included.**



- **Activity - Do It:**
  - **<u>Materials Needed</u>**

🎬 Arduino board (e.g., Arduino Uno)

🎬 USB cable for Arduino

🎬 Small submersible water pump (e.g., 5V)

🎬 Relay module (a single-channel relay module compatible with Arduino) - *Note: A relay is needed because the pump likely requires more current/voltage than the Arduino pins can directly provide.*

🎬 Push button or toggle switch

🎬 Breadboard

🎬 Jumper wires

🎬 Small container of water

🎬External power supply for the water pump (if the pump requires more than the Arduino can safely provide, check pump specifications)

🎬Computer with Arduino IDE installed

- **Step-by-Step Instructions**

🎬**Introduction to Components:** Briefly introduce the function of each component: Arduino (the brain), Switch (input), Water Pump (output), and Relay (the bridge allowing Arduino to control the higher-power pump).
🎬**Circuit Building - Switch:** Guide students to connect the push button switch to the Arduino. Typically, one side of the switch connects to a digital pin on the Arduino, and the other side connects to Ground (GND), often with a pull-up or pull-down resistor (internal or external) to ensure a stable reading. *Provide a simple circuit diagram.*
🎬**Circuit Building - Relay:** Explain the relay module. Guide students to connect the relay's control pins (usually VCC, GND, and Signal/IN) to the Arduino (VCC to 5V, GND to GND, Signal to a digital pin).
🎬**Circuit Building - Water Pump:** Guide students to connect the water pump to the relay's screw terminals. The relay acts like an electronic switch for the pump's power supply. Connect the pump's power supply (positive terminal) to the 'Common' (COM) terminal on the relay and the pump's positive wire to either the 'Normally Open' (NO) or 'Normally Closed' (NC) terminal (NO is common for turning something on when the relay is activated). Connect the pump's negative wire to the ground of its power supply. *Emphasize double-checking connections before applying power.*
🎬**Initial Code Attempt:** Have students open the Arduino IDE. Guide them to write basic code:

- Define pin numbers for the switch and relay.
- In `setup()`, configure the switch pin as an INPUT (with `INPUT_PULLUP` if using internal pull-up) and the relay pin as an OUTPUT.
- In `loop()`, read the state of the switch using `digitalRead()`.
- Use an `if` statement to check if the switch is pressed.
- If the switch is pressed, use `digitalWrite()` to activate the relay (this might be HIGH or LOW depending on the relay module).
- If the switch is not pressed, use `digitalWrite()` to deactivate the relay.

🎬**Testing:**

- Upload the code to the Arduino.
- Place the water pump in the container of water.
- Connect the external power supply for the pump (if using one).
- Test the circuit by pressing the switch. Observe if the pump turns on and off as expected.

🎬**Troubleshooting:** Encourage students to identify issues (e.g., pump not turning on, pump always on). Guide them to check:

- Wire connections (are they secure and in the correct pins/terminals?).
- Code logic (is the `if` statement correct? Is the relay activated by HIGH or LOW?).
- Power supply for the pump.

- **Reflection Questions**

🎬What did you connect to the Arduino inputs and outputs?

🎬How did the switch tell the Arduino what to do?

🎬How did the Arduino tell the water pump (through the relay) what to do?

🎬What challenges did your group face while building the circuit or writing the code? How did you try to solve them?

🎬If you wanted the pump to stay on for a specific amount of time after pressing the button, how might you change the code?

🎬How is controlling this water pump similar to or different from controlling the water pump on a real fire truck (based on your "Engage" discussion)?

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4. Explain

- **Teacher Role:**

  Concept Clarifier: Explain complex ideas and introduce relevant vocabulary (e.g., electromagnetism, contacts, coil, NO, NC, COM).
  Discussion Leader: Facilitate a class discussion to ensure understanding and address misconceptions.
  Connector: Help students connect what they learned from the video and discussion back to their hands-on experience in the Explore phase.
  Assessor (Formative): Gauge student understanding through their explanations and participation in discussions.

- **Expected From Students:**

  Active Listening/Watching: Pay attention to the video and teacher explanations.
  Participate in Discussion: Share their understanding, ask questions, and respond to prompts.
  Explain Concepts: Attempt to explain how a relay works in their own words.
  Make Connections: Relate the video content to their Arduino circuit and the role of the relay in their project.
  Vocabulary Acquisition: Learn and use new terms related to relays and circuits.

- **Parts of the books included.**

- **Watch It:**

Pre-Watching: Give students specific questions to think about or look for while watching the video. Examples:

- "What are the main parts inside a relay?"
- "How does the small signal from the Arduino control the larger power for the pump?"
- "Why can't we connect the pump directly to the Arduino?

During Watching:
🎬 **Show the Video:** Play a clear, age-appropriate educational video explaining how a relay works. (Teacher will need to select a suitable video beforehand).

🎬 **Pause and Discuss (Optional but Recommended):** Pause the video at key points to check for understanding, clarify difficult concepts

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Post-Watching:

🎬**Class Discussion:** Facilitate a deeper discussion about the video content.

- "Based on the video, can someone explain in their own words how the relay works?"
- "What does the small current from the Arduino do inside the relay?"
- "How does that then control the larger current for the pump?"
- "Why was the relay so important in our fire truck pump circuit?"
- "What does 'Normally Open' (NO) and 'Normally Closed' (NC) mean on the relay terminals?" (Connect to how they wired their pump).

🎬**Connect to Explore Activity:** Explicitly link the video explanation back to their hands-on experience. "Now that you've seen how a relay works, does it make sense why you connected the wires the way you did in the Explore activity?"

Implementation Strategies:

1. Visual Cues: Pause the video to draw students' attention to critical components of the circuit.

2. Question Prompts: Use targeted questions during pauses to encourage engagement and deepen understanding.

3. Class Discussion: Summarize the video as a class and use a diagram on the board to reinforce the concepts learned

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5. Elaborate

- **Teacher Role:**

  Guide and Facilitator: Provide guidance on integration strategies and materials, but allow students to make design decisions.
  Problem-Solving Support: Assist students in troubleshooting issues that arise during the integration process.
  Encourager of Creativity: Promote innovative solutions for mounting components and designing the fire truck's functionality.
  Assessor (Formative): Observe how students apply their understanding and skills in a new context. Ask questions to prompt deeper thinking about their design choices.
  Safety Supervisor: Ensure safe use of tools and materials during the construction/integration process.

- **Expected From Students:**

  Design and Plan: Think about the best way to arrange and attach the components to their fire truck model.

Integrate: Physically connect and secure the Arduino, relay, pump, and switch onto the fire truck model.
Connect Systems: Attach tubing or a "hose" from the water pump outlet to a suitable point on the truck.
Test and Refine: Test the integrated system (pump water from the truck) and make adjustments as needed.
Problem Solve: Identify and overcome challenges related to mounting, wiring, and water flow within the context of the model.
Collaborate: Work with teammates to decide on the design and share tasks.

- **<u>Parts of the books included.</u>**

- **Practice.**

Materials Needed:

🎬Your working Arduino pump system (from the last activity).

🎬A model fire truck (can be made from recycled materials).

🎬Things to attach parts: Tape, glue, zip ties, etc.

🎬Small tube for the water hose.

🎬Basic tools: Scissors, maybe a craft knife (use carefully!).

🎬Water.

Step-by-Step Instructions:

🎬**Plan:** Decide where each part (Arduino, pump, etc.) will go on your truck.

🎬**Attach Parts:** Securely stick or tie the parts onto the truck.

🎬**Add the Hose:** Connect the small tube to the pump and run it to where the water should come out on the truck.

🎬**Check Wires:** Make sure wires are neat and won't get caught.

🎬**Test:** Put the pump in water and see if it works when you use the switch.

🎬**Fix Problems:** If it doesn't work right, try to figure out why and fix it.

Reflection Questions:

🎬 Why did you put the parts where you did?

🎬 What was hard about putting everything on the truck? How did you fix it?

🎬 Does your fire truck pump water like you expected? How could you make it better?

🎬 How did thinking about controlling the pump earlier help you now?

🎬 What did you learn about building electronic things onto a model?

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6. Evaluate

- **Teacher Role:**

Assessor: Evaluate student understanding based on their responses to the questions.
Feedback Provider: Offer constructive feedback to students based on their performance.
Instructional Adjuster: Identify areas where students struggled and plan for reteaching or further practice if necessary.
Data Collector: Use student responses to inform future instruction.

Expected Student:

- **Demonstrate Understanding:** Answer questions accurately based on their learning throughout the 5E cycle.
- **Recall Information:** Remember key facts and concepts about relays and water pumps.
- **Apply Knowledge:** Use their understanding to select the correct answers.
- **Communicate Learning:** Articulate their understanding through their chosen responses.

- **Parts of the books included.**

Focus

1. What is the primary function of a relay in an Arduino project?

A) To increase the speed of the Arduino's processor.
B) To control high-power devices with low-power Arduino outputs.
C) To store data in non-volatile memory.
D) To improve the Arduino's wireless communication range.

2. The main purpose of a water pump is to:

A) Move water from a higher level to a lower one.
B) Move water from a lower level to a higher one.
C) Compress water.
D) Transfer water between two points and remove excess water.

3. What type of energy conversion occurs in water pumps?

A) Thermal energy into electrical energy.
B) Mechanical energy into chemical energy.
C) Electrical energy into mechanical energy.
D) Kinetic energy into potential energy.

Possible Answers:

Based on the questions provided in the image, the correct answers are:

1. **B) To control high-power devices with low-power Arduino outputs.** (This aligns with the explanation of how relays protect the Arduino and control components like motors/pumps).
2. **B) Move water from a lower level to a higher one.** (While pumps can transfer water between any two points, their primary function often involves overcoming gravity or pressure differences, typically moving from lower to higher potential energy or pressure zones).
3. **C) Electrical energy into mechanical energy.** (Water pumps use electrical energy to power a motor, which then performs mechanical work to move the water).

Strategies for Implementation:

- **Quiz:** Use these questions as a short quiz to individually assess student understanding.
- **Exit Ticket:** Pose one or two questions as an exit ticket at the end of the lesson or unit.

Project Rubric: Arduino Water Pump Fire Truck

This rubric can be used to evaluate student groups on their work in building and integrating the Arduino-controlled water pump system into their fire truck model.

Criteria	Beginning (1)	Developing (2)	Proficient (3)	Advanced (4)	Score
Circuit Construction	Many incorrect connections; circuit does not work.	Some incorrect connections; circuit works inconsistently.	Circuit is mostly correct; works with minor issues.	Circuit is correctly wired and connections are secure.	

Arduino Code	Code is missing or has major errors; does not control pump.	Code has errors; controls pump inconsistently or partially.	Code is mostly correct; controls pump with minor bugs.	Code is correct, well-structured, and controls the pump reliably.	
Integration into Truck	Components are not attached or poorly placed; hose not connected.	Components are attached but placement is awkward; hose connection is loose.	Components are reasonably well-attached and placed; hose connected.	Components are securely and thoughtfully integrated; hose is well-connected and routed.	
Functionality	Pump does not turn on/off with the switch.	Pump turns on/off inconsistently or requires adjustment.	Pump turns on/off correctly most of the time.	Pump turns on/off reliably and effectively with the switch.	
Troubleshooting & Problem-Solving	Unable to identify or fix problems.	Identifies some problems but struggles to find solutions.	Identifies most problems and attempts solutions with some success.	Effectively identifies problems and implements successful solutions.	
Collaboration (if applicable)	Group struggles to work together; tasks are not shared.	Group works together sometimes but faces challenges; some tasks shared.	Group works together effectively most of the time; tasks are generally shared.	Group collaborates effectively; tasks are clearly shared, and members support each other.	
Overall Design & Creativity	Little evidence of planning or creative design.	Some attempt at design; basic integration.	Design is functional; integration is practical.	Design is creative and well-executed; integration is thoughtful and enhances the model.	

Total Score: ________ / 28

Notes:

- Adjust criteria and scoring based on the specific age and experience level of your students.
- Consider adding a section for student reflection or presentation of their work.
- This rubric can be used for formative assessment during the Elaborate phase or for summative assessment upon completion of the project.

Showcase: Evaluate the students work in the main project and add a rubric so teachers can follow .

❖ **Relation to Standards:**

[**Science Standards (like NGSS - Next Generation Science Standards)**

- **Building and Fixing Things (Engineering Design):** This is a big part of the lesson!
  - Students work to solve a real problem: making a fire truck pump water (Engage, Elaborate).
  - They plan, build, and test their system (Explore, Elaborate).
  - They figure out what works and how to make it better (Explore, Elaborate).
- **Energy:**
  - Students learn that electric power can be changed into movement power to make the pump work (Explain - energy change).
  - They use electric power to make their fire truck model do something (Elaborate).

Engineering and Technology Standards (like ISTE Standards)

- **Using Tools to Create (Innovative Designer):**
  - Students use tools like Arduino and electronic parts to design and build their project (Explore, Elaborate).
  - They create a working system to solve a problem (simulating fire fighting) (Elaborate).
- **Thinking Like a Computer (Computational Thinker):**
  - Students learn to think step-by-step to make the pump work with code (Explore).
  - They write simple code and fix any problems they find (Explore, Elaborate).

21st Century Skills

This lesson helps students practice skills important for school and life:

- **Collaboration:** Students work together in groups to build, code, and troubleshoot (Explore, Elaborate).
- **Critical Thinking & Problem-Solving:** Students have to figure out why their circuit or code isn't working and find ways to fix it (Explore, Elaborate). They also think about the best way to design their fire truck integration (Elaborate).
- **Creativity & Innovation:** Students can be creative in how they design and build their fire truck model and integrate the system (Elaborate).
- **Communication:** Students talk with their group members about their ideas and challenges, and explain what they learned (All Phases, especially Explain and Evaluate).

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Challenge questions of the lesson:

☐ Can you make the buzzer sound *different* when the flame is closer or farther away?

☐ Can you add a *light* that turns on when the alarm sounds?

☐ Can you make the alarm sound only after the flame has been there for a few seconds?

UNIT 2: [Name of the unit]

Chapter 01: [Robotics domain]

Lesson 2: [evacuate]

Time Frame: [90 min]

SEL Relation

□ **Showing things:** They can see the parts (sensor, buzzer, Arduino) and how they connect.
□ **Using pictures:** Diagrams help them understand the wiring.
□ **Working in groups:** They can talk and learn words from friends while building.
□ **Doing things:** Connecting wires and seeing the buzzer sound helps them understand words like "connect," "sensor," "sound," and "alarm" by doing.
□ **Showing what they know:** They can show their working alarm even if they can't explain everything perfectly in English.

1. **Think**

- **Questions**

- **Possible Answers**

Possible Student Answers: Students might come up with a variety of ideas, drawing on their experiences. Possible answers could include:

- Something that smells smoke.
- Something that feels heat.
- Something that sees fire (like a camera).
- It needs a loud sound (a buzzer, a siren).
- It needs a light that flashes.
- It needs power (batteries, electricity).
- It needs to be put in the right place (on the ceiling, in rooms).

- **Strategies to implement:**

Brainstorming/Think-Pair-Share:

- Give students a moment to think quietly about the question.
- Have them pair up to share their ideas with a partner.
- Bring the class back together to share ideas from their pairs. Record all responses on a board or chart paper.

2. Explore

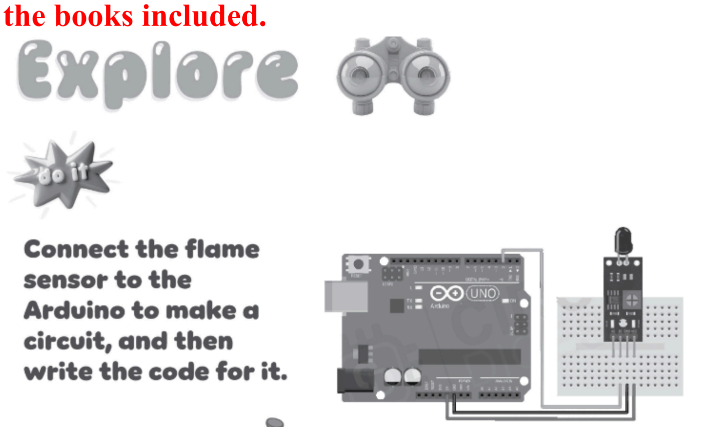
Teacher Role:

- **Guide:** Help students build the circuit and write code.
- **Watch:** See how students are working and if they understand.
- **Help Fix Problems:** Guide students when their circuit or code doesn't work.
- **Keep Everyone Safe:** Make sure students are very careful, especially if using a small flame for testing (teacher should handle any flame source).

Expected Student Actions:

- **Work Together:** Build the circuit with a partner or group.
- **Connect Wires:** Follow steps to hook up the sensor to the Arduino.
- **Write Code:** Try writing or changing code to read the sensor.
- **Test It:** See if the sensor works by testing it carefully.
- **Fix Problems:** Try to find and fix mistakes in wires or code.
- **See What Happens:** Watch what the sensor tells the Arduino.

Parts of the books included.



Materials Needed (per group):

- Arduino board

- USB cable
- Flame sensor module
- Jumper wires
- Computer with Arduino IDE
- **For Testing (Teacher Only):** A small, safe flame source (like a candle in a holder), used ONLY by the teacher and with extreme caution.

Step-by-Step Instructions:

1. **Look at the Sensor:** Find the pins on the flame sensor (usually labeled VCC, GND, and maybe DO/AO).
2. **Connect Power:** Connect the VCC pin on the sensor to 5V on the Arduino. Connect the GND pin on the sensor to GND on the Arduino.
3. **Connect Signal:** Choose a digital pin on the Arduino (like pin 7) and connect it to the digital output (DO) pin on the flame sensor. (Optional: Connect the analog output (AO) to an analog pin like A0 for more detailed readings).
4. **Open Arduino IDE:** Start the coding program on the computer.

5. **Write Simple Code:**
    - Tell the Arduino which pin the sensor is on.
    - In the `setup` part, tell the Arduino to use the Serial Monitor to show numbers.
    - In the `loop` part, read the signal from the sensor pin.
    - Print the sensor reading to the Serial Monitor.
    - Wait a little bit before reading again.
6. **Send Code to Arduino:** Connect the Arduino to the computer and upload the code.
7. **Open Serial Monitor:** Click the Serial Monitor button in the Arduino IDE to see the numbers.
8. **Test Carefully (Teacher does this part):** The teacher will safely bring a small flame near the sensor.
9. **Watch the Numbers:** See how the numbers in the Serial Monitor change when the flame is near or far.

Reflection Questions:

- What numbers did you see in the computer when there was NO flame?
- What numbers did you see when a flame was NEAR the sensor?
- How does the sensor know a flame is close?
- How could these changing numbers help us make a fire alarm sound?
- What was tricky about connecting the sensor or writing the code?

3. Explain

Teacher Role:

- **Concept Clarifier:** Explain how flame sensors detect fire and introduce related vocabulary (e.g., infrared light, sensor, signal, threshold).
- **Discussion Leader:** Guide a class discussion to make sure students understand the video and the sensor.
- **Connector:** Help students link what they learn from the video to their experience using the sensor in the Explore phase.
- **Assessor (Formative):** Check student understanding through their answers and participation.

Expected Student Actions:

- **Watch and Listen:** Pay close attention to the video and teacher's explanations.
- **Talk About It:** Share their ideas and ask questions during the discussion.
- **Explain How it Works:** Try to describe how the flame sensor detects a flame using new words they learned.

- **Make Connections:** See how the video explains the numbers they saw in the Serial Monitor during the Explore activity.

Parts of the books included

Strategies for Implementation ("Watch It" - Understanding the Flame Sensor):

1. **Pre-Watching:**
   - **Connect to Explore:** Start by asking about their experience in the last activity. "You connected a flame sensor and saw numbers change. What did you notice about the numbers when a flame was near?"
   - **Introduce the Video:** Tell students they will watch a video to learn *how* the flame sensor actually "sees" a flame.
   - **What to Look For:** Give them things to listen or watch for: "How does the sensor detect fire?", "What does the signal or number from the sensor tell us?", "Why is this sensor good for a fire alarm?"

2. **During Watching:**
   - **Show the Video:** Play an educational video that explains how flame sensors work, perhaps showing the part that detects light. (Teacher will need to choose a good video beforehand).
   - **Pause (Optional):** Stop the video at important parts to ask questions or explain things more simply.

3. **Post-Watching:**
   - **Check Understanding:** Ask the questions you gave them before the video. Discuss their answers as a class.
   - **Class Discussion:** Talk more about the video.
     - "What kind of light does the sensor look for?" (Infrared)
     - "How does the sensor send a message to the Arduino?" (Through a signal/number)
     - "What did the different numbers in the Serial Monitor mean?" (More flame = different number/signal)
     - "How is this sensor different from a smoke detector?" (Detects flame light vs. smoke particles)
   - **Link to Explore:** Ask, "Now that you've seen the video, does it help explain why the numbers changed when you put a flame near your sensor?"
   - **Introduce Vocabulary:** Use words like "sensor," "detect," "signal," "infrared light," and "threshold" (the point where the signal means "fire").

5.Elaborate

Teacher Role:

- **Guide Integration:** Help students correctly add the buzzer to their existing circuit.
- **Support Coding:** Assist students in modifying their code to control the buzzer based on the sensor reading.
- **Help Troubleshoot:** Guide students in finding and fixing problems in the combined circuit and code.
- **Connect to Real World:** Discuss how this simple system is like a real alarm.

Expected Student Actions:

- **Add Component:** Connect the buzzer to the Arduino circuit.
- **Update Code:** Change the existing code to make the buzzer work with the sensor.
- **Test the Alarm:** See if the buzzer sounds when a flame is detected and turns off when it's not.
- **Fix Problems:** Work together to find and fix any issues.
- **Collaborate:** Work effectively with their group members.

Parts of the books included

Materials Needed (per group):

- Their working Arduino and flame sensor circuit from the Explore phase.
- A buzzer (most simple buzzers have 2 pins).
- Jumper wires.
- Computer with Arduino IDE.
- **For Testing (Teacher Only):** A small, safe flame source (like a candle in a holder), used ONLY by the teacher and with extreme caution.

Step-by-Step Instructions:

1. **Look at the Buzzer:** Find the pins on the buzzer (usually marked + and - or just two pins).
2. **Connect Buzzer:** Connect the positive (+) pin of the buzzer to a digital pin on the Arduino (like pin 8). Connect the negative (-) pin of the buzzer to GND on the Arduino.
3. **Open Code:** Open the flame sensor code you used before in the Arduino IDE.
4. **Add Buzzer Code:**
   - At the top, tell the Arduino which pin the buzzer is on.
   - In the `setup` part, tell the Arduino that the buzzer pin is for output.
   - In the `loop` part, find where you read the flame sensor.
   - Add an `if` statement: `if` the flame sensor reading means a flame is detected (you'll need to figure out the right condition based on your sensor's output from the Explore phase), then turn the buzzer pin HIGH to make sound.
   - Add an `else` statement: `else` (if no flame is detected), turn the buzzer pin LOW to turn off the sound.
5. **Send Code to Arduino:** Upload the new code to the Arduino.
6. **Test the Alarm (Teacher does this part):** The teacher will safely bring a small flame near the sensor.
7. **Listen:** See if the buzzer sounds when the flame is near and stops when it's taken away.

Reflection Questions:

- How did you decide *when* the buzzer should turn on in your code?

- What does the Arduino do when the flame sensor detects a flame?
- How is this circuit like a simple fire alarm?
- What was the hardest part about adding the buzzer or changing the code?
- What else could you add to make this a better fire alarm? (Like a light, or sending a message).

6. Evaluate

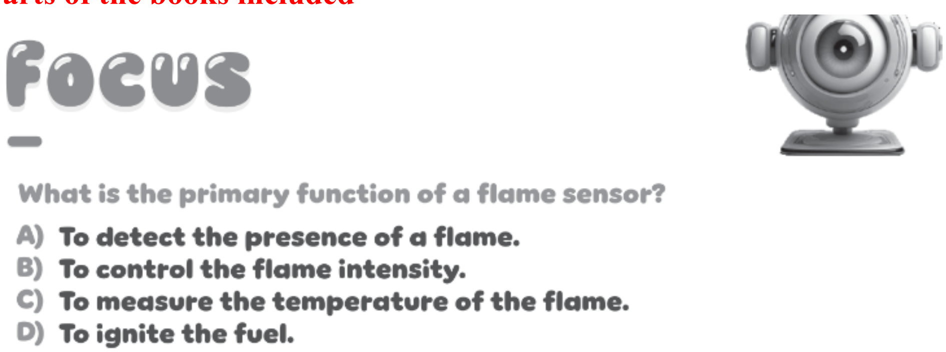
Teacher Role:

- **Assess Understanding:** Use student responses to see if they understand what a flame sensor does.
- **Identify Needs:** Figure out if students need more help or review on this topic.
- **Give Feedback:** Provide feedback on their answer.

Expected Student Actions:

- **Show Learning:** Answer the question based on what they learned in the lesson.
- **Recall Information:** Remember the main job of the flame sensor.

Parts of the books included


Focus

What is the primary function of a flame sensor?

A) To detect the presence of a flame.
B) To control the flame intensity.
C) To measure the temperature of the flame.
D) To ignite the fuel.

Possible Answer:

Based on the question and the lesson's focus, the correct answer is:
A) To detect the presence of a flame. (The lesson focused on the sensor identifying if a flame is there).

Strategies for Implementation:

- **Quick Check:** Use this as a quick question at the end of the lesson.
- **Exit Ticket:** Have students write their answer on a small piece of paper before leaving.
- **Show of Hands:** Read the options and have students raise their hand for the answer they think is correct.

Project Rubric: Arduino Water Pump Fire Truck

This rubric can be used to evaluate student groups on their work in building and integrating the Arduino-controlled water pump system into their fire truck model.

Criteria	Beginning (1)	Developing (2)	Proficient (3)	Advanced (4)	Score
Circuit Construction	Many incorrect connections; circuit does not work.	Some incorrect connections; circuit works inconsistently.	Circuit is mostly correct; works with minor issues.	Circuit is correctly wired and connections are secure.	
Arduino Code	Code is missing or has major errors;	Code has errors; controls pump	Code is mostly correct; controls	Code is correct, well-structured, and	

	does not control pump.	inconsistently or partially.	pump with minor bugs.	controls the pump reliably.	
Integration into Truck	Components are not attached or poorly placed; hose not connected.	Components are attached but placement is awkward; hose connection is loose.	Components are reasonably well-attached and placed; hose connected.	Components are securely and thoughtfully integrated; hose is well-connected and routed.	
Functionality	Pump does not turn on/off with the switch.	Pump turns on/off inconsistently or requires adjustment.	Pump turns on/off correctly most of the time.	Pump turns on/off reliably and effectively with the switch.	
Troubleshooting & Problem-Solving	Unable to identify or fix problems.	Identifies some problems but struggles to find solutions.	Identifies most problems and attempts solutions with some success.	Effectively identifies problems and implements successful solutions.	
Collaboration (if applicable)	Group struggles to work together; tasks are not shared.	Group works together sometimes but faces challenges; some tasks shared.	Group works together effectively most of the time; tasks are generally shared.	Group collaborates effectively; tasks are clearly shared, and members support each other.	
Overall Design & Creativity	Little evidence of planning or creative design.	Some attempt at design; basic integration.	Design is functional; integration is practical.	Design is creative and well-executed; integration is thoughtful and enhances the model.	

Total Score: ________ / 28

Notes:

- Adjust criteria and scoring based on the specific age and experience level of your students.
- Consider adding a section for student reflection or presentation of their work.
- This rubric can be used for formative assessment during the Elaborate phase or for summative assessment upon completion of the project.

Science Standards (like NGSS - Next Generation Science Standards)

- **Building and Fixing Things (Engineering Design):** This is a big part of the lesson!
  - Students work to solve a real problem: making a fire truck pump water (Engage, Elaborate).
  - They plan, build, and test their system (Explore, Elaborate).
  - They figure out what works and how to make it better if needed (Explore, Elaborate).
- **Energy:**
  - Students see how electric power can be changed into movement power to make the pump work (Explain - energy change).
  - They use electric power to make their fire truck model do something (Elaborate).

Engineering and Technology Standards (like ISTE Standards)

- **Using Tools to Create (Innovative Designer):**

    - Students use tools like Arduino and electronic parts to design and build their project (Explore, Elaborate).
    - They create a working system to solve a problem (simulating fire fighting) (Elaborate).
- **Thinking Like a Computer (Computational Thinker):**
    - Students learn to think step-by-step to make the pump work with code (Explore).
    - They write simple code and fix any problems they find (Explore, Elaborate).

21st Century Skills
This lesson helps students practice skills important for school and life:
- **Collaboration:** Students work together in groups to build, code, and troubleshoot (Explore, Elaborate).
- **Critical Thinking & Problem-Solving:** Students have to figure out why their circuit or code isn't working and find ways to fix it (Explore, Elaborate). They also think about the best way to design their fire truck integration (Elaborate).
- **Creativity & Innovation:** Students can be creative in how they design and build their fire truck model and integrate the system (Elaborate).
- **Communication:** Students talk with their group members about their ideas and challenges, and explain what they learned (All Phases, especially Explain and Evaluate).

UNIT 3:

Chapter 01: [programming domain]
Lesson 01: [control the speed truck]
Time Frame: [90 min]

Objectives:

- Control the speed of the fire truck through app inventor.
- Make an app for emergency call.
- Understand the function of clock, texting and location sensor block.

Challenge questions of the lesson:

- Can you make the phone app control the fire truck's *lights* too?
- Can you make the truck move in a *circle* with one button press?
- Can you make the truck stop automatically if it gets too close to something (using another sensor)?

SEL Relation

- **Showing things:** They can see the Bluetooth part, the phone app, and the truck moving.
- **Using pictures:** The App Inventor blocks use pictures to show code.
- **Working in groups:** They can talk and learn words from friends while building and coding.
- **Doing things:** Connecting the Bluetooth, sending messages, and seeing the truck move helps them understand words like "control," "wireless," "send," "receive," and "move" by doing.
- **Showing what they know:** They can show their working app and truck even if they can't explain everything perfectly in English.

1.Engage

Teacher Role:

- **Introduce the New Challenge:** Present the idea of controlling the fire truck remotely.
- **Spark Curiosity:** Make students wonder how a phone can "talk" to a fire truck.
- **Connect to Real Life:** Relate controlling the fire truck to other things students might control with phones or remotes.
- **Listen to Ideas:** Encourage students to share their initial thoughts and questions.

Expected Student Actions:

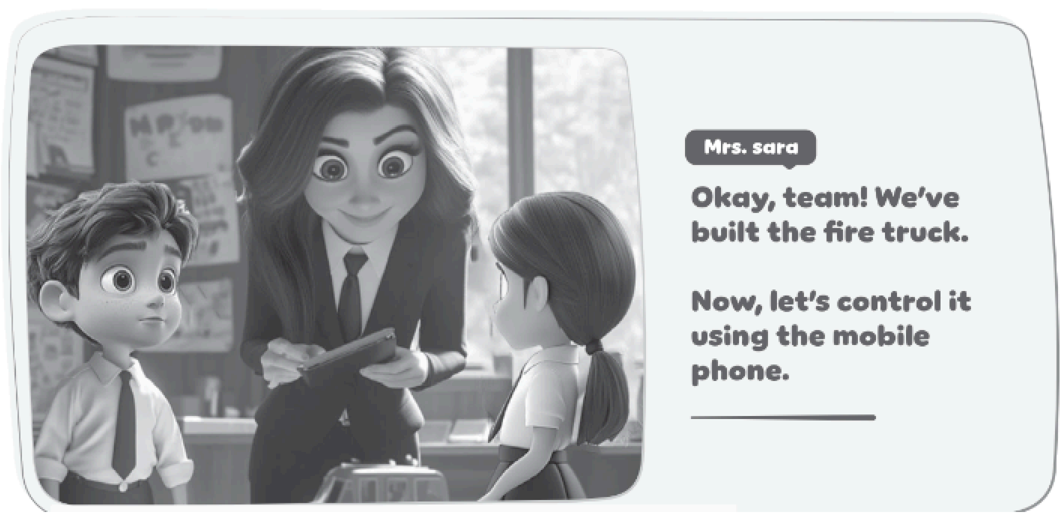
🎬 **Listen and Think:** Pay attention to the new challenge presented by Mrs. Sara.

🎬 **Share Ideas:** Talk about things they already control with phones or remotes.

🎬 **Ask Questions:** Wonder how a phone could control a physical model.

🎬 **Make Connections:** Relate the idea to their own experiences with technology.

Parts of the books included.



- **Strategies implemented:**

Visual or Multimedia Engagement:

🎬 **Open Discussion:** Ask students:

- "What is Mrs. Sara suggesting they do now?"
- "Have you ever controlled something with a mobile phone or a remote control?" (Think about TVs, toys, lights, etc.)
- "How do you think a mobile phone could control our fire truck model?" (Allow for all ideas - magic, signals, etc.)
- "Why might controlling a fire truck with a phone be useful?"

🎬 **Pose a Question for Thought:** Ask, "What kind of 'message' do you think the phone needs to send to the fire truck to tell it what to do?"

2.Think

- **Questions**

- **Possible Answers**

Students might suggest upgrades based on their understanding of the system so far. Possible ideas could include:

- **Better Sensors:** A sensor that can tell how *big* the fire is, not just if it's there. A sensor that can tell *how far away* the fire is.
- **Automatic Control:** Make the pump turn on by itself when the fire alarm goes off, instead of waiting for a person to press a button.
- **Adjustable Water Pressure:** Control how strong the water comes out based on the fire size.
- **Remote Aiming:** Control the direction of the water hose from the phone.
- **Battery Level Sensor:** Add a way to know how much power the truck has left.
- **GPS/Location:** Know exactly where the truck is or where the fire is.

- **Strategies to implement:**

Brainstorming/Think-Pair-Share (Efficiency Upgrades):

- Give students a moment to think quietly about the question. Consider what "efficiency" means in this context (faster response, less wasted water, more accurate control, etc.).
- Have them pair up to share their ideas with a partner.
- Bring the class back together to share ideas from their pairs. Record all responses on a board or chart paper.

3.Explore

Teacher Role:

🎬 **Provide Code:** Give students the Arduino code file that works with a mobile app slider.

🎬 **Guide Code Review:** Help students look through the code to understand how it reads Bluetooth data and processes potential slider values.

🎬 **Facilitate Testing:** Guide students in connecting the phone app to the Bluetooth module and using the slider control.

- **Support Troubleshooting:** Help students fix issues with code upload, Bluetooth connection, or receiving data.

- **Connect Observations:** Help students link what they see in the Serial Monitor to the movement of the slider on the phone.

Expected Student Actions:

- **Get Code:** Obtain the Arduino code file.

- **Review Code:** Read through the code with their group, trying to understand what it does.

- **Upload Code:** Connect the Arduino (with the Bluetooth module attached) to the computer and upload the code.

- **Connect Bluetooth:** Use the mobile app to connect to the Bluetooth module.

- **Test Slider:** Use the slider control in the mobile app.

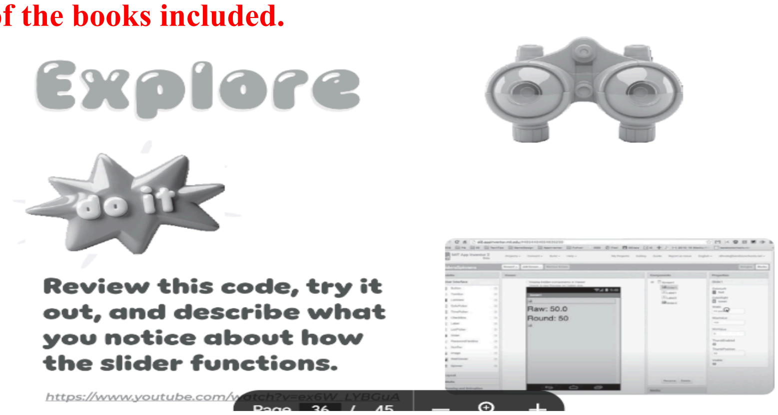
- **Observe Data:** Watch the Serial Monitor in the Arduino IDE to see what data is sent by the slider.

- **Describe Findings:** Explain what they noticed about how the numbers or messages in the Serial Monitor changed as they moved the slider.

- **Fix Problems:** Work together to troubleshoot any issues.

- **Collaborate:** Work effectively with teammates.

Parts of the books included.



Materials Needed (per group):

- Arduino board

- USB cable

🎬 HC-05 or HC-06 Bluetooth module (connected to Arduino - *refer to previous connection steps if needed*)

🎬 Jumper wires (if reconnecting module)

🎬 Computer with Arduino IDE

🎬 A mobile phone (or tablet) with a Bluetooth control app that includes a slider (e.g., "Bluetooth Serial Controller" or a custom app).

🎬 **The Arduino code file for the slider function (provided by the teacher).**

Step-by-Step Instructions:

- 🎬 **Get the Code:** Find the Arduino code file provided by your teacher.
- 🎬 **Open in Arduino IDE:** Open the code file using the Arduino program on the computer.
- 🎬 **Look at the Code:** Read through the code. Try to find parts that read information from Bluetooth and parts that might use a number from the slider.
- 🎬 **Connect Arduino:** Make sure your Arduino is connected to the computer with the USB cable. (Make sure the Bluetooth module is connected to the Arduino correctly - VCC to 5V, GND to GND, TX to Arduino RX pin, RX to Arduino TX pin).
- 🎬 **Upload Code:** Click the upload button in the Arduino IDE to send the code to the Arduino.
- 🎬 **Open Serial Monitor:** Click the Serial Monitor button in the Arduino IDE.
- 🎬 **Connect Phone App:** On your phone, open the Bluetooth control app and connect it to your Bluetooth module.

- 🎬 **Use the Slider:** Find the slider control in the app and move it back and forth.
- 🎬 **Watch and Write:** Look at the numbers or messages that appear in the Serial Monitor. Write down what you notice about how the numbers change when you move the slider.
- What numbers did you see in the computer when there was NO flame?
- What numbers did you see when a flame was NEAR the sensor?
- How does the sensor know a flame is close?
- How could these changing numbers help us make a fire alarm sound?
- What was tricky about connecting the sensor or writing the code?

Reflection Questions:

- What kind of information (numbers or text) did the slider send to the Arduino?
- How did moving the slider from one side to the other change the information the Arduino received?
- If the slider went from 0 to 255, what number did you see when the slider was in the middle?
- How do you think the Arduino code uses the number from the slider?

- How could we use a slider like this to control something on our fire truck, like the speed of a motor or the strength of the water pump?

4. Explain

Teacher Role:

🎬 **Concept Clarifier:** Explain how a mobile app sends data, how the Arduino receives it, and how that data can be used to control speed (e.g., faster number = faster speed). Introduce vocabulary (e.g., App Inventor, interface, data, control signal, speed control).

🎬 **Discussion Leader:** Guide a class discussion to make sure students understand the steps shown in the video and the overall process.

🎬 **Connector:** Help students link what they see in the video (sending data from the app) to their experience receiving slider data in the Explore phase.

🎬 **Assessor (Formative):** Check student understanding through their answers and participation.

Expected Student Actions:

🎬 **Watch and Listen:** Pay close attention to the video, especially how the app is built and how it sends data.

🎬 **Talk About It:** Share their ideas and ask questions during the discussion.

🎬 **Explain the Process:** Try to describe the steps from moving the slider on the phone to the Arduino potentially changing a motor's speed.

🎬 **Make Connections:** See how the App Inventor part connects to the Bluetooth and Arduino code they explored earlier.

Parts of the books included
Strategies for Implementation ("Watch It" - Understanding the Flame Sensor):

🎬 **Pre-Watching:**

- **Connect to Explore:** Start by referring back to the Explore activity. "You saw that the slider on the phone app sent different numbers to the Arduino. How do you think those numbers could tell the fire truck how fast to go?"
- **Introduce the Video:** Tell students they will watch a video that shows how to build the mobile app side using a tool called App Inventor and how that app sends the slider number to control speed.
- **What to Look For:** Give them specific things to watch for: "How is the slider added to the app?", "How does the app send the slider's number using Bluetooth?", "How does the code on the Arduino use that number to control speed?"

🎬 **During Watching:**

- **Show the Video:** Play an educational video demonstrating how to create a simple app with a slider in App Inventor that sends the slider value via Bluetooth, and potentially shows the Arduino code receiving this value and using it for speed control (e.g., using `analogWrite()` for PWM if controlling a DC motor). (Teacher will need to choose a good video beforehand).
- **Pause (Optional):** Stop the video at key points to explain the App Inventor interface, the Bluetooth blocks in App Inventor, or the part of the Arduino code that uses the received number to set speed.

🎬 **Post-Watching:**

- **Check Understanding:** Ask the questions you gave them before the video. Discuss their answers as a class.
- **Class Discussion:** Talk more about the video.
  - "What tool was used to build the mobile app?" (App Inventor)
  - "How did the app know what number to send when the slider moved?"
  - "How did the app send the number wirelessly?" (Using Bluetooth blocks/commands)
  - "Once the Arduino gets the number, how does it use it to control speed?" (Explain that a higher number can mean more power or faster movement).
  - "How does this connect to the numbers you saw in the Serial Monitor in the last activity?"
- **Link to Explore:** Ask, "Now that you've seen how the app sends the number, does it make more sense how your Arduino was receiving those different values when you moved the slider?"

5.Elaborate
Teacher Role:

🎬 **App Inventor Guide:** Introduce students to the App Inventor website and guide them through designing the app interface and programming the blocks to send Bluetooth commands.

🎬 **Coding Support:** Assist students in writing or modifying the Arduino code to receive and interpret commands sent from the app (e.g., parsing text commands and speed values).

🎬 **Integration Facilitator:** Help students connect motors and a motor driver to the Arduino and integrate them into the fire truck model (if not already done).

🎬 **Troubleshooting Expert:** Guide students in diagnosing and fixing problems across the entire system: the app, the Bluetooth connection, the Arduino code, the motor wiring, and the motor driver.

🎬 **Project Manager:** Help groups manage their time and tasks for this multi-part project.

🎬 **Celebrator:** Acknowledge and celebrate student successes as they get parts of their system working.

Expected Student Actions:

🎬 **Design App Interface:** Plan and create the look of their control app in App Inventor (buttons for directions, slider for speed).

🎬 **Program App Logic:** Use App Inventor blocks to make the buttons and slider send specific commands or values via Bluetooth when used.

🎬 **Write/Modify Arduino Code:** Write or update the Arduino code to receive the Bluetooth data, understand the commands (like "Forward" or "Turn Left"), and read the speed value from the slider.

🎬 **Control Motors:** Write code to use the motor driver to make the motors spin in the correct direction and at the speed received from the app.

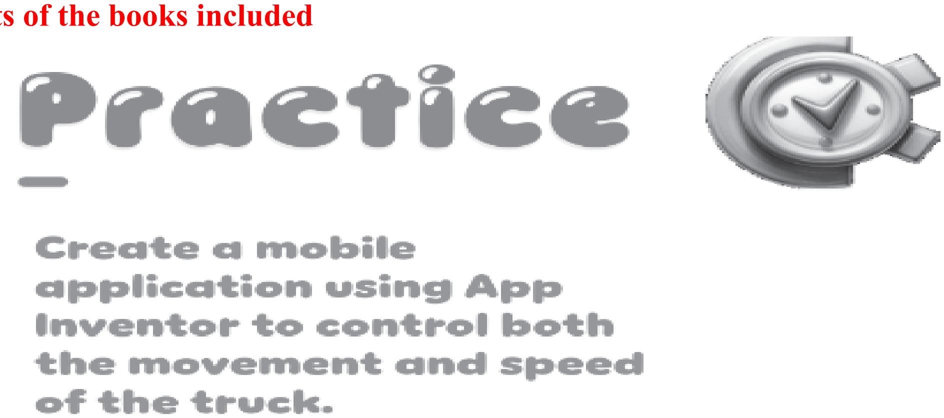
🎬 **Integrate Hardware:** Connect the motors and motor driver to the Arduino and the fire truck model (if necessary).

🎬 **Test the System:** Test the app controls to see if the fire truck moves correctly and changes speed.

🎬 **Troubleshoot:** Identify and fix problems in the app, Bluetooth connection, Arduino code, or motor wiring.

🎬 **Collaborate:** Work effectively as a team to complete all parts of the project.

Parts of the books included



Materials Needed (per group):

🎬 Their fire truck model (ideally already built and possibly with motors/mounting points).

🎬 Arduino board

🎬 USB cable

🎬 HC-05 or HC-06 Bluetooth module (connected to Arduino).

🎬 Jumper wires.

🎬 Motor driver module (like L298N) - *Needed to control DC motors with Arduino.*

🎬 DC Motors (2 or 4, depending on how the truck moves).

🎬 Separate power supply for motors (e.g., AA batteries in a battery holder) - *Motors need more power than Arduino can give.*

🎬 Computer with internet access (to use App Inventor online) and Arduino IDE installed.

🎬 Mobile phone or tablet with the MIT App Inventor Companion app installed.

🎬 Basic tools for integration (tape, glue, screwdriver, etc.).

Step-by-Step Instructions:

🎬 **Plan Your App:** On paper, draw what your control app screen will look like. What buttons do you need (Forward, Back, Left, Right)? Where will the speed slider go?

🎬 **Build the App Interface (App Inventor):** Go to the App Inventor website. Start a new project. Drag and drop buttons, a slider, and a Bluetooth client component onto your screen design. Make them look nice.

🎬 **Program the App Blocks (App Inventor):** Switch to the Blocks editor. Use the Bluetooth blocks to connect to the Bluetooth module. Program the buttons to send specific letters or words (like "F" for Forward). Program the slider to send its number value when it's moved. You might send a letter command followed by the slider number (e.g., "F150").

🎬 **Prepare Arduino Code:** Open your Arduino IDE. You'll need code to:

- Set up communication with the Bluetooth module (like you did in Explore).
- Set up the pins for the motor driver.
- In the `loop` part, read the commands coming from Bluetooth.
- Use `if` statements to check which command was received (like "F").
- Read the speed number that comes after the command.
- Tell the motor driver to make the motors move in the right direction and at the speed from the slider number.

🎬 **Connect Motors and Driver:** If your truck model doesn't have motors and a driver, connect the motors to the motor driver module. Connect the motor driver's control pins to the Arduino. Connect the motor driver to its separate battery power supply. Connect the motor driver's ground to the Arduino's ground.

🎬 **Integrate Hardware:** Attach the motors, motor driver, Arduino, and Bluetooth module to your fire truck model. Make sure wires are neat and secure.

🎬 **Upload Arduino Code:** Connect the Arduino to the computer and upload your code.

🎬 **Test the System:**

- Make sure the motor batteries are connected.
- Open your app on the phone and connect to the Bluetooth module.
- Try pressing the buttons and moving the slider.
- See if the fire truck moves and changes speed.

🎬 **Troubleshoot and Refine:** If it doesn't work, check: Is the app connected? Is the Bluetooth module powered? Is the Arduino code running? Are the motor driver and motors wired correctly? Is the motor battery charged? Is the code reading the commands and speed correctly? Fix problems and try again.

Reflection Questions:

🎬 How did you design your app screen? Why did you choose those buttons or controls?

🎬 How does your app tell the Arduino to move the truck forward or backward? How does it control the speed?

🎬 What was the hardest part about building the app or writing the Arduino code for this project?

🎬 How well does your phone control the fire truck's movement and speed?

- If you had more time, what other controls or features would you add to your app or your fire truck?
- How is building this app and controlling the truck like real-world remote control technology?

6. **Evaluate**

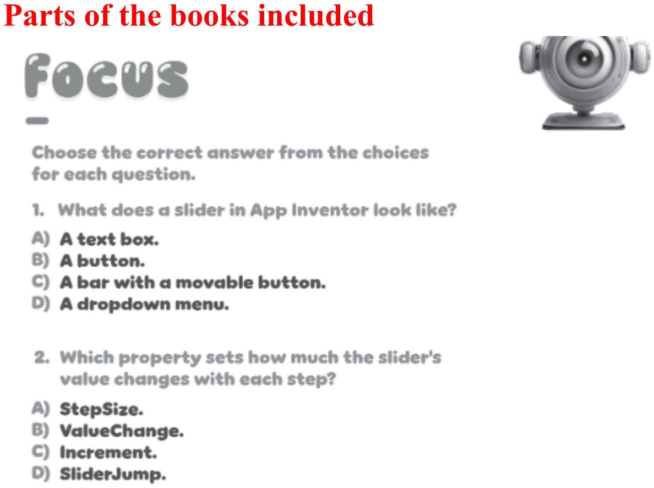
Teacher Role:

- **Assess Understanding:** Use student responses to see if they understand what a slider looks like and how its properties work in App Inventor.
- **Identify Needs:** Figure out if students need more help or review on using App Inventor components.
- **Give Feedback:** Provide feedback on their answers.

Expected Student Actions:

- **Show Learning:** Answer the questions based on what they learned about App Inventor sliders.
- **Recall Information:** Remember the visual appearance and properties of the slider component.

Parts of the books included



Focus

Choose the correct answer from the choices for each question.

1. What does a slider in App Inventor look like?

A) A text box.
B) A button.
C) A bar with a movable button.
D) A dropdown menu.

2. Which property sets how much the slider's value changes with each step?

A) StepSize.
B) ValueChange.
C) Increment.
D) SliderJump.

Possible Answer:

Based on the questions provided in the image and standard App Inventor components:

1. **C) A bar with a movable button.** (This is the typical visual representation of a slider).
2. **A) StepSize.** (In App Inventor, the `ThumbPosition` property changes, and `StepSize` can influence how the value changes when the thumb is dragged or clicked, though the primary property for tracking the value is `ThumbPosition` itself. However, among the options provided, `StepSize` is the most relevant property related to how the value *could* change in steps if programmed that way or if the slider has specific step behavior enabled).

Strategies for Implementation:

- **Quick Check:** Use these questions as a quick check after students have worked with the slider in App Inventor.
- **Exit Ticket:** Have students write their answers on a small piece of paper before leaving.

🎬 **Individual Response:** Students write their chosen letter for each question.

Project Rubric: Arduino Water Pump Fire Truck

This rubric can be used to evaluate student groups on their work in building and integrating the Arduino-controlled water pump system into their fire truck model.

Criteria	Beginning (1)	Developing (2)	Proficient (3)	Advanced (4)	Score
Circuit Construction	Many incorrect connections; circuit does not work.	Some incorrect connections; circuit works inconsistently.	Circuit is mostly correct; works with minor issues.	Circuit is correctly wired and connections are secure.	
Arduino Code	Code is missing or has major errors; does not control pump.	Code has errors; controls pump inconsistently or partially.	Code is mostly correct; controls pump with minor bugs.	Code is correct, well-structured, and controls the pump reliably.	
Integration into Truck	Components are not attached or poorly placed; hose not connected.	Components are attached but placement is awkward; hose connection is loose.	Components are reasonably well-attached and placed; hose connected.	Components are securely and thoughtfully integrated; hose is well-connected and routed.	
Functionality	Pump does not turn on/off with the switch.	Pump turns on/off inconsistently or requires adjustment.	Pump turns on/off correctly most of the time.	Pump turns on/off reliably and effectively with the switch.	
Troubleshooting & Problem-Solving	Unable to identify or fix problems.	Identifies some problems but struggles to find solutions.	Identifies most problems and attempts solutions with some success.	Effectively identifies problems and implements successful solutions.	
Collaboration (if applicable)	Group struggles to work together; tasks are not shared.	Group works together sometimes but faces challenges; some tasks shared.	Group works together effectively most of the time; tasks are generally shared.	Group collaborates effectively; tasks are clearly shared, and members support each other.	
Overall Design & Creativity	Little evidence of planning or creative design.	Some attempt at design; basic integration.	Design is functional; integration is practical.	Design is creative and well-executed; integration is thoughtful and enhances the model.	

Total Score: ________ / 28

Notes:

- Adjust criteria and scoring based on the specific age and experience level of your students.
- Consider adding a section for student reflection or presentation of their work.

- This rubric can be used for formative assessment during the Elaborate phase or for summative assessment upon completion of the project.

Science Standards (like NGSS - Next Generation Science Standards)

- **Building and Fixing Things (Engineering Design):** This is a big part of the lesson!
    - Students work to solve a real problem: making a fire truck pump water (Engage, Elaborate).
    - They plan, build, and test their system (Explore, Elaborate).
    - They figure out what works and how to make it better if needed (Explore, Elaborate).
- **Energy:**
    - Students see how electric power can be changed into movement power to make the pump work (Explain - energy change).
    - They use electric power to make their fire truck model do something (Elaborate).

Engineering and Technology Standards (like ISTE Standards)

- **Using Tools to Create (Innovative Designer):**
    - Students use tools like Arduino and electronic parts to design and build their project (Explore, Elaborate).
    - They create a working system to solve a problem (simulating fire fighting) (Elaborate).
- **Thinking Like a Computer (Computational Thinker):**
    - Students learn to think step-by-step to make the pump work with code (Explore).
    - They write simple code and fix any problems they find (Explore, Elaborate).

21st Century Skills

This lesson helps students practice skills important for school and life:

- **Collaboration:** Students work together in groups to build, code, and troubleshoot (Explore, Elaborate).
- **Critical Thinking & Problem-Solving:** Students have to figure out why their circuit or code isn't working and find ways to fix it (Explore, Elaborate). They also think about the best way to design their fire truck integration (Elaborate).
- **Creativity & Innovation:** Students can be creative in how they design and build their fire truck model and integrate the system (Elaborate).
- **Communication:** Students talk with their group members about their ideas and challenges, and explain what they learned (All Phases, especially Explain and Evaluate).

UNIT 3:

Chapter 01: [robotics domain]
Lesson 02: [SOS!]
Time Frame: [90 min]

Objectives:

- Explain why location is important in an emergency app.
- Design a simple app screen for an SOS button.
- Use App Inventor to get the phone's location and time.
- Use App Inventor to send a message with information.
- Work with friends to build an app.

SEL Relation

- **Showing things:** They can see the app design and blocks, not just hear words.
- **Using pictures:** App Inventor uses pictures (blocks) for coding.
- **Working in groups:** They can talk and learn words from friends.
- **Doing things:** Building the app helps them understand words like "button," "send," and "location" by doing the actions.
- **Showing what they know:** They can show their working app even if they can't explain everything perfectly in English.

Challenge questions of the lesson:

- Can you make the app send the message to a *real* phone number?
- Can you add a place for the user to type *what kind* of emergency it is?
- Can you make the app send the message automatically if you shake the phone?

1.Engage

Teacher Role:

- **Introduce the Problem:** Present the real-world problem of needing to quickly alert emergency services.

- **Highlight the Solution Idea:** Draw attention to Laila's idea of a mobile SOS application.

- **Spark Curiosity:** Make students wonder how a phone app could send a message to firefighters.

- **Connect to Real Life:** Discuss existing emergency alert systems or ways people call for help.

- **Listen to Ideas:** Encourage students to share their initial thoughts and questions about SOS apps.

Expected Student Actions:

- **Listen and Think:** Pay attention to the new challenge presented in the story.
- **Share Ideas:** Talk about ways people get help in emergencies.
- **Ask Questions:** Wonder about how a mobile app could contact emergency services.
- **Make Connections:** Relate the idea to emergency numbers they know (like 999 or 112)

Parts of the books included.

- **Strategies implemented:**

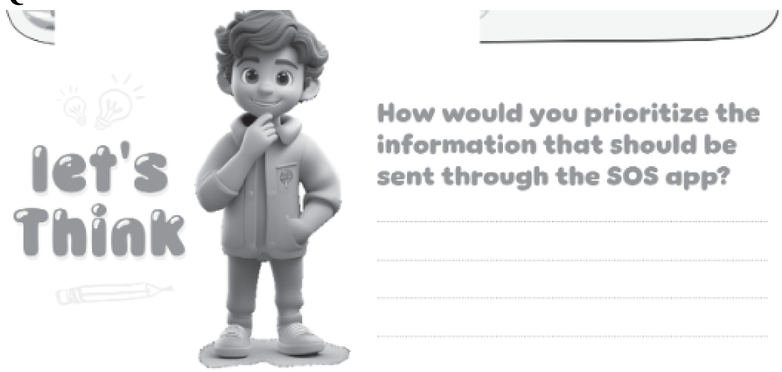
Visual or Multimedia Engagement:

- **Present the Hook:** Read or show the story snippet with Laila and Adam's idea.
- **Open Discussion:** Ask students:
  - "What problem are Laila and Adam trying to solve?"
  - "What is their idea for solving it?"
  - "Why is it important to contact firefighters *immediately* in an emergency?"
  - "How do people usually call for help in an emergency?" (Discuss calling emergency numbers).
  - "How do you think a mobile app could send a message or alert to firefighters?" (Allow for all initial ideas).

🎬 **Pose a Question for Thought:** Ask, "What information would an SOS app need to send to be helpful to firefighters?" (Guide thinking towards location, type of emergency, etc.)

2.Think

- **Questions**



- **Possible Answers**

Students might suggest different pieces of information and reasons for their priority. Possible ideas could include:

- **Location:** (Highest Priority) - "Firefighters need to know *where* to go first!" (Address, GPS coordinates).
- **Type of Emergency:** (High Priority) - "Is it a fire, a medical problem, or something else?" (Simple category).
- **Your Name/Contact Info:** (High Priority) - "So they can call you back for more details."
- **Number of People Involved/Trapped:** (High Priority) - "This helps them know how many people might need help."

- **Severity of Emergency:** (Medium/High Priority) - "Is it a small fire or a big one?" (Maybe a simple rating or description).
- **Details of the Situation:** (Medium Priority) - "What is on fire? Are there any dangers like chemicals?" (More detailed description).
- **Your Status:** (Lower Priority for initial message, but helpful) - "Are you safe? Are you injured?"

- **Strategies to implement:**

Brainstorming/Think-Pair-Share (Efficiency Upgrades):

Brainstorming/Think-Pair-Share (Prioritizing Information):

- Give students a moment to think quietly about the question. What information is absolutely essential? What can wait? What would be helpful but maybe not critical for the first message?
- Have them pair up to share their ideas with a partner, discussing which pieces of information are most important to send *first*.
- Bring the class back together to share ideas from their pairs. Record all responses on a board or chart paper, perhaps grouping similar ideas and discussing why certain information is more critical than others.

3.Explore

Teacher Role:

🎬 **Introduce Design:** Explain what a user interface is and why good design is important for an app, especially an emergency one.

🎬 **Introduce the Tool:** Show students how to access and use the online sketching tool (Excalidraw or similar).

🎬 **Guide Design:** Help students think about what buttons, text areas, or other elements are needed on the app screen based on their "Think" discussion about important information.

🎬 **Support Tool Use:** Assist students as they use the sketching software.

🎬 **Facilitate Sharing:** Create opportunities for students to share their design ideas with others.

Expected Student Actions:

🎬 **Brainstorm UI Elements:** Think about what needs to be on the app screen (buttons, places for text, etc.).

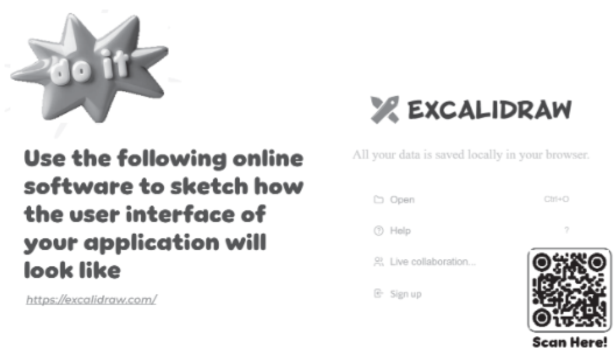
🎬 **Use Sketching Tool:** Learn and use the online software to draw their app screen design.

🎬 **Design the Layout:** Decide where to place elements on the screen.

🎬 **Collaborate:** Work with their group to create a shared design.

🎬 **Explain Design:** Be ready to talk about why they designed the app screen the way they did.

Parts of the books included.



Materials Needed (per group):

🎬 Computer or tablet with internet access.

🎬 Access to an online sketching tool like Excalidraw (https://excalidraw.com/).

🎬 Paper and pencil for initial ideas (optional)

Step-by-Step Instructions:

🎬 **Think About the App:** Remember our discussion about what information is most important to send in an SOS message (like location, type of emergency).

🎬 **What Goes on the Screen?** What buttons or places to type information do you need on your app screen so someone can quickly send an SOS? (Example: A big SOS button, maybe a place to quickly select "Fire" or "Medical").

🎬 **Go to the Website:** Open the online sketching tool (like Excalidraw) on your computer.

🎬 **Start Drawing:** Use the tools to draw shapes for your phone screen and the different parts of your app screen (buttons, boxes for text).

🎬 **Add Labels:** Write text on your drawing to show what each button or area is for (like "SOS," "My Location," "Type of Emergency").

🎬 **Arrange Everything:** Move the parts around to make the app screen easy to use, especially in an emergency.

🎬 **Save or Share:** Save your sketch or be ready to show it to the class.

Reflection Questions:

🎬 Why did you put the SOS button in that place on the screen?

🎬 What is the most important thing on your app screen? Why?

🎬 How did you decide what other information to include on the screen besides the SOS button?

🎬 Would your app be easy to use quickly in an emergency? Why or why not?

🎬 What was easy or hard about designing the app screen?

5. Explain

Teacher Role:

🎬 **Concept Clarifier:** Explain what GPS and location services are in simple terms. Explain the function of the Location Sensor and Clock components in App Inventor. Introduce vocabulary (e.g., GPS, coordinates, latitude, longitude, timestamp, sensor, properties, events).

🎬 **Discussion Leader:** Guide a class discussion to ensure students understand how the components work and why they are important for an SOS app.

🎬 **Connector:** Help students link what they see in the video to their design ideas from the Explore phase and the need for location information discussed in the Engage phase.

🎬 **Assessor (Formative):** Check student understanding through their answers and participation.

Expected Student Actions:

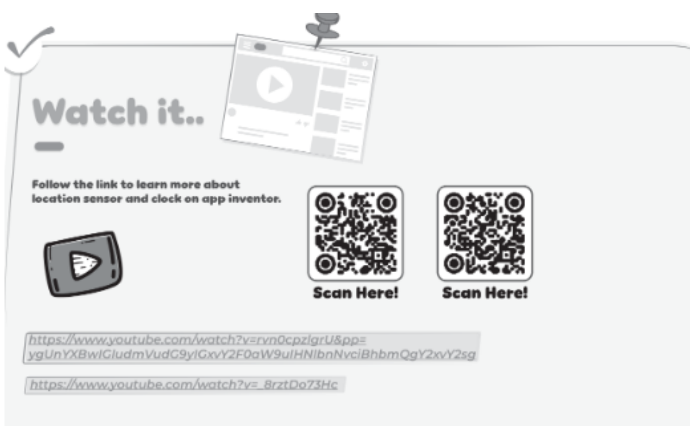
🎬 **Watch and Listen:** Pay close attention to the video, especially how the Location Sensor and Clock components are used in App Inventor.

🎬 **Talk About It:** Share their ideas and ask questions during the discussion.

🎬 **Explain How it Works:** Try to describe how the app gets the location and uses the clock.

🎬 **Make Connections:** See how these components are necessary for sending important information in an emergency.

Parts of the books included



Strategies for Implementation ("Watch It" - Understanding the Flame Sensor):

🎬 **Pre-Watching:**

🎬 **Connect to Engage/Explore:** Start by referring back to the "Think" discussion in the Engage phase about prioritizing information. "We said that knowing the *location* is one of the most important things in an SOS message. How do you think a phone app can find out where it is?"

🎬 **Introduce the Video:** Tell students they will watch a video that shows how to use special parts in App Inventor (called components) to get the phone's location and the current time.

🎬 **What to Look For:** Give them specific things to watch for: "How does the app get the location?", "What kind of information does the location sensor give us?", "What does the clock component do?", "How can these help our SOS app?"

🎬 **During Watching:**

> 🎬 **Show the Video:** Play an educational video demonstrating how to add and use the Location Sensor and Clock components in App Inventor. The video should show how to get latitude, longitude, and potentially a timestamp. (Teacher will need to choose a good video beforehand).
>
> 🎬 **Pause (Optional):** Stop the video at key points to explain concepts like latitude and longitude, how the sensor needs permission, or how the clock can provide the current time.

🎬 **Post-Watching:**
🎬 **Check Understanding:** Ask the questions you gave them before the video. Discuss their answers as a class.
🎬 **Class Discussion:** Talk more about the video.

- "What is the name of the component that finds the phone's location?" (Location Sensor)
- "What two main numbers does the location sensor give us to know where we are?" (Latitude and Longitude - explain these are like coordinates on a map).
- "What is the Clock component used for?" (Getting the current time, maybe setting timers).
- "Why is getting the location so important for an SOS app?"
- "How could knowing the exact time (using the Clock) be helpful in an emergency message?" (Knowing *when* the emergency happened).

🎬 **Link to Explore:** Ask, "Now that you've seen how to get the location, where on the app screen design you sketched could you show the user their location?"
🎬 **Introduce Vocabulary:** Use words like "Location Sensor," "GPS," "latitude," "longitude," "coordinates," "Clock," "timestamp."

5.Elaborate
Teacher Role:

🎬 **App Inventor Guide:** Help students navigate the App Inventor website, add components, design the screen, and program the blocks.

🎬 **Concept Connector:** Remind students of the "Think" discussion about important information (like location) and how the Location Sensor and Clock can provide this (from the "Explain" phase).

🎬 **Coding Support:** Assist students in using the blocks editor to get location/time data and send a message using the phone's capabilities.

🎬 **Troubleshooting Expert:** Help students fix issues with the app design, blocks programming, or testing on their phone.

🎬 **Safety Reminder:** Emphasize responsible use of the app and not sending fake emergency messages.

Expected Student Actions:

🎬 **Design App:** Create the layout of the SOS app screen in App Inventor.
🎬 **Add Components:** Add necessary components like a Button, Location Sensor, Clock, and a way to send messages (like the Sharing component or Activity Starter for sending texts).
🎬 **Program Blocks:** Use App Inventor blocks to:

- Get the current location (latitude and longitude) when the app starts or a button is pressed.
- Get the current time.
- Put the location and time information into a message.
- Send the message when the SOS button is clicked.

🎬 **Test the App:** Connect their phone and test if the button sends a message with the correct information.
🎬 **Troubleshoot:** Identify and fix problems in their app design or blocks.
🎬 **Collaborate:** Work together in their group to build the app.

Parts of the books included

Materials Needed (per group):

🎬 Computer with internet access (to use App Inventor online).

🎬 Mobile phone or tablet with the MIT App Inventor Companion app installed.

🎬 Access to the internet on the mobile device (for location services and potentially sending messages).

🎬 App Inventor account (free).

Step-by-Step Instructions:

🎬 **Go to App Inventor:** Open the App Inventor website on your computer.

🎬 **Start a New Project:** Create a new project for your SOS app.

🎬 **Design the Screen:**

- Add a big Button on the screen (maybe label it "SOS" or "Send Help").
- Add a Label to show the location (latitude and longitude).
- Add a Label to show the time.
- Add a **LocationSensor** component (find it under "Sensors"). This component is invisible on the screen.
- Add a **Clock** component (find it under "Sensors"). This component is also invisible.
- Add a way to send a message. You can use the **Sharing** component (under "Social") to share text, or the **ActivityStarter** component (under "Connectivity") to open the phone's messaging app with a pre-filled message. *Choose one method based on what you want the app to do.*

🎬 **Go to Blocks:** Click the "Blocks" button to start programming.

🎬 **Get Location and Time:**

- Use blocks for the **LocationSensor** to get the current Latitude and Longitude. You might do this when the screen opens or when the SOS button is clicked.
- Use blocks for the **Clock** to get the current time (like `Clock.Now`).

🎬 **Build the Message:** Use "join" blocks (under "Text") to put together a message string. Include text like "SOS! Emergency at:" followed by the Latitude and Longitude, and maybe "Time:" followed by the time.

🎬 **Program the SOS Button:**

- Find the block for `when Button1.Click do`.
- Inside this block, add the blocks to get the location and time, build the message string, and then use the blocks for your chosen messaging component (Sharing or ActivityStarter) to send the message.

🎬 **Test on Your Phone:**

- Open the App Inventor Companion app on your phone.
- In App Inventor on the computer, go to "Connect" and choose "AI Companion."
- Scan the QR code with your phone.
- Your app should appear on the phone.
- Press the SOS button to see if it tries to send a message with your location and time.

🎬 **Troubleshoot and Refine:** If it doesn't work, check: Did you add all the components? Are the blocks connected correctly? Is your phone connected to the internet and does it have location services turned on? Did you grant location permission to the App Inventor app?

Reflection Questions:

🎬 What information does your SOS app send when you press the button?

🎬 How did you get the app to find your location?

🎬 How did you get the app to know the time?

🎬 What was the hardest part about building this app in App Inventor?

🎬 How could you make this SOS app even better or more helpful in a real emergency? (Think about sending it to a specific number, adding more details, etc.)

6. **Evaluate**

Teacher Role:

🎬 **Assess Understanding:** Use student responses to see if they understand the purpose of the App Inventor components covered.

🎬 **Identify Needs:** Figure out if students need more help or review on using these specific blocks/components.

🎬 **Give Feedback:** Provide feedback on their answers.

Expected Student Actions:

🎬 **Show Learning:** Answer the questions based on what they learned about the App Inventor components.

🎬 **Recall Information:** Remember the functions of the Clock, Location Sensor, and Texting blocks.

Parts of the books included

Focus

1. Which block would you use to schedule an event to occur every 5 seconds?

A) Texting.
B) Clock.
C) LocationSensor.
D) Timer.

2. What information does the LocationSensor NOT provide?

A) Latitude.
B) Longitude.
C) Altitude.
D) Weather.

3. How can you send a text message to a specific phone number in App Inventor?

A) Use the Clock block.
B) Use the Texting Send Message block.
C) Use the LocationSensor block.
D) Use the GetLatitude block.

Possible Answer:

Based on the questions provided in the image and standard App Inventor components:

1. **B) Clock.** (The Clock component has timer events that can be set to trigger at intervals, like every 5 seconds).
2. **D) Weather.** (The Location Sensor provides geographical coordinates like Latitude, Longitude, and Altitude, but not weather information).
3. **B) Use the Texting Send Message block.** (The Texting component in App Inventor is specifically designed for sending SMS messages).

Strategies for Implementation:

🎬 **Quick Check:** Use these questions as a quick check after students have completed the Elaborate activity.

🎬 **Exit Ticket:** Have students write their answers on a small piece of paper before leaving.

🎬 **Individual Response:** Students write their chosen letter for each question.

🎬 **Review Questions:** Discuss these questions as a class to reinforce understanding of the components.

Project Rubric: Arduino Water Pump Fire Truck

Science Standards (like NGSS - Next Generation Science Standards)

- **Gathering Information:** Students learn how a phone app can collect information from sensors, like finding its location using GPS (Explain - Location Sensor).
- **Using Data:** Students learn that the app needs to use the location and time data to create a helpful message (Elaborate).

Engineering and Technology Standards (like ISTE Standards)

- **Building Apps (Engineering Design):**
    - Students work to solve a real problem: how to quickly ask for help in an emergency (Engage).
    - They design how the app will look and work (Explore).
    - They build the app using a tool (App Inventor) and test it (Elaborate).
    - They figure out how to make the app send the right information (Elaborate).
- **Using Tools to Create (Innovative Designer):**
    - Students use a special online tool (App Inventor) to design and build their app (Explore, Elaborate).
    - They create a useful product (the SOS app) to help in emergencies (Elaborate).
- **Thinking Like a Computer (Computational Thinker):**
    - Students plan the steps the app needs to follow (like get location, get time, build message, send message) (Elaborate - Blocks).
    - They use blocks in App Inventor to tell the app what to do (programming) (Elaborate).
    - They find and fix mistakes in their app's code (Elaborate).

21st Century Skills

This lesson helps students practice skills important for school and life:

- **Collaboration:** Students work together in groups to design and build the app (Explore, Elaborate).
- **Critical Thinking & Problem-Solving:** Students think about what information is most important (Engage - Think). They figure out how to get that information and how to make the app work correctly (Elaborate - Troubleshooting).
- **Creativity & Innovation:** Students design their app screen and decide how it will work (Explore, Elaborate). They create a new tool to solve a problem.
- **Communication:** Students talk with their group members about their ideas and challenges (All Phases). They design the app to communicate important information to others (Elaborate).